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St. Joseph's college of Arts & Science (Autonomous)

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B. Sc DEGREE EXAMINATION-APRIL2021

BC102-CELL BIOLOGY

Time: 3 hrs **Max Marks: 75**

SECTION-A (25X1=25)

Answer ALL Questions

I. Choose the correct answer: (10X1=10)

1. Protoplasm found inside the nucleus is known as

   (a)Amyloplast (b) Cytoplasm

   (c) Nucleoplasm (d) Elaioplasm

2. Nucleus was discovered by

   (a) Leeuwenhoek (b) Robert Hooke

   (c) Robert Brown (d) Robert Swanson

3. Which one of the following organelles digest the old organelles that is no longer useful to the cells

   (a) Ribosomes (b) Mitochondria

   (c) Lysosomes (d) Chromatin

4. Carrier proteins transport single solute from one side of the membrane to the other

   (a) Uniport (b) Symport

   (c) Antiport (d) Cotransport

5. Glyoxysome found in _____

   (a) Plant cells (b) Animal cells

   (c) Viruses (d) All the above

6. Animal cell differs from plant cells in possessing
   (a) Plastid (b) Golgi body
   (c) Vacuole (d) Centrosome
7. Microsomes which lack ribosomes are called
   (a) Rough Endoplasmic Reticulum
   (b) Smooth Endoplasmic Reticulum
   (c) Granular epithelium
   (d) Glycosomes
8. Which one of the following is not considered as a part of the endomembrane system
   (a) Vacuole (b) Lysosome
   (c) Golgi complex (d) Peroxisome
9. According to the fluid mosaic model of cell membranes, which type of molecule spans the membrane, from its inner to outer surface?
   (a) phospholipid (b) protein
   (c) carbohydrate (d) cholesterol
10. Microfilaments play main role in
   (a) Cytokinesis
   (b) Chromosome movement
   (c) Intracellular movement of cellular material
   (d) Contraction of bones

II Say true or false (5X1=5)

11. No metabolic energy is consumed in passive transport.
12. Meiosis would be used in asexual reproduction.

13. The cytoplasm of a prokaryotic cell contains the cell's DNA.
14. Chromosomes with the same order of genes are called homologous.
15. The form of ubiquinones present in mitochondria is often called coenzymeQ_{10}

III Answer all the questions (10X1=10)

16. Define endocytosis.
17. Define prokaryotic cells.
18. Define enzyme.
19. What are ribosomes?
20. What is microbody?
21. Define cell cycle.
22. Define chromatin.
23. What are microtubules?
24. What are IF proteins?
25. Define peroxisomes.

SECTION-B (4X5=20)

Answer any FOUR of the following:

26. Compare the prokaryotic and eukaryotic cell.
27. Explain the structure of fluid mosaic model.
28. Describe structure and functions of mitochondria
29. Explain morphology and functions of Golgi complex
30. Explain the phases of cell cycle
31. Write a note on microtubules and microfilaments.

SECTION-C (3X10=30)

Answer any THREE of the following:

32. What is membrane transport system? Explain passive and active transport system.
33. Explain morphology and functions of Endoplasmic reticulum.
34. Give an account of ribosomes structure and functions
35. Explain mitotic cell cycle with labeled sketch.
36. Discuss components of cytoskeleton and its biochemical functions.

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